

Method	Success $\uparrow$	Cost $\downarrow$	Route $\uparrow$	Collision $\downarrow$
Baseline	0.000	1.000	0.190	0.627
Curriculum	0.000	1.000	0.177	0.627
Risk-only	0.000	0.000	0.009	0.000
No guard	0.000	1.000	0.172	0.513
Guard	0.633	0.207	0.868	0.200
Shield	0.627	0.227	0.871	0.227
Full	0.620	0.247	0.865	0.233
Retuned full	0.620	0.240	0.853	0.220
Gated risk	0.600	0.253	0.849	0.253

Table 1: Formal held-out evaluation at density 0.15 with three-seed means. Seed-level variability is reported in the robustness appendix; zero-cost risk-only results are not interpreted as safety because route progress collapses.

1 Experimental Protocol

All formal experiments use the frozen 16-environment protocol recorded in the experiment ledger. Training and evaluation are performed with `n_envs=16` unless an experiment is explicitly marked as diagnostic. The formal held-out densities are 0.00, 0.08, and 0.15, with 50 evaluation episodes per density and per seed. The high-density main table reports density 0.15, where unsafe shortcuts and no-progress policies are most visible. Stress tests at densities 0.20 and 0.25 are frozen-model robustness probes rather than tuning targets.

The comparison set separates mechanism from incidental reward tuning. Negative controls include vanilla PPO, curriculum-only PPO, risk-only PPO, and a no-action-guard ablation. Positive mechanism controls include `guard_only` and `shield_only`; full, retuned, and gated-risk variants test whether dense risk shaping, TTC, lane, smoothness, and conditional risk gating add robustness beyond the action-level safety mechanism. All paper-facing tables are generated from the frozen paper-table package and indexed by the final table manifest.

2 Results

2.1 Formal Held-Out Comparison

Table 1 shows that dense-traffic performance is dominated by the action-level safety mechanism rather than by risk reward alone. Vanilla PPO and curriculum-only PPO both fail at density 0.15, with zero success and unit cost. The no-action-guard ablation also collapses despite sharing the surrounding framework. Risk-only PPO has zero success and nearly zero route completion, so its zero cost is a stagnation failure rather than safe driving.

The strongest formal rows are the guard/shield-centered variants. `guard_only` reaches a success rate of 0.633 with cost 0.207 and route completion 0.868, while `shield_only` reaches 0.627 success with cost 0.227 and route completion 0.871. Full and gated variants remain competitive, but they do not consistently exceed the simpler action-guard baselines. The central manuscript claim should therefore be guard/shield-centered robustness under dense traffic, with risk shaping and gated-risk variants framed as refinements.

2.2 Component Ablation

Table 2 gives the mechanism ablation at density 0.15. Removing the action guard is the most damaging intervention: success falls to zero and cost rises to 1.000. Removing the TTC term degrades high-density

Method	Success $\uparrow$	Cost $\downarrow$	Route $\uparrow$	Collision $\downarrow$
No guard	0.000	1.000	0.172	0.513
w/o TTC	0.540	0.360	0.823	0.360
w/o lane	0.607	0.280	0.844	0.280
w/o smooth	0.567	0.287	0.821	0.280
Guard	0.633	0.207	0.868	0.200

Table 2: Mechanism ablation at density 0.15. Removing the action guard collapses held-out performance, while TTC, lane, and smoothness terms act as refinements around the guard-centered mechanism.

Density	Rank	Method	Success $\uparrow$	Cost $\downarrow$	Route $\uparrow$
0.20	1	Shield	0.480	0.360	0.778
0.20	2	Retuned full	0.453	0.380	0.753
0.20	3	Gated risk	0.447	0.367	0.766
0.20	4	Guard	0.413	0.407	0.733
0.20	5	Full	0.380	0.400	0.744
0.20	6	Risk-only	0.000	0.000	0.009
0.25	1	Guard	0.267	0.487	0.658
0.25	2	Gated risk	0.220	0.527	0.628
0.25	3	Full	0.207	0.587	0.587
0.25	4	Shield	0.193	0.587	0.609
0.25	5	Retuned full	0.173	0.687	0.571
0.25	6	Risk-only	0.000	0.000	0.009

Table 3: Stress evaluation at densities 0.20 and 0.25, ranked by success, cost, and route completion. These rows are robustness evidence, not tuning objectives.

safety, with success 0.540 and cost 0.360. Removing the lane term increases lateral deviation and worsens the cost-success tradeoff, while removing smoothness mainly affects stability and seed variance.

These ablations show that the method is not a single reward-weight trick. The action guard provides the dominant stabilizing structure, and the reward terms refine collision anticipation, lateral stability, and control smoothness around that structure.

2.3 Stress-Density Robustness

Table 3 evaluates frozen models at densities 0.20 and 0.25. These densities are harder than the formal held-out table and should not be described as training objectives. At density 0.20, **shield_only** ranks first by success/cost/route ordering, while **proposed_gated_risk** and the retuned full variant remain close. At density 0.25, **guard_only** becomes the cleanest survivor among progress-making methods. Risk-only again shows near-zero route completion and should be interpreted as stagnation rather than safety.

The stress results support graceful degradation of guard/shield-centered policies. They do not support a claim that full risk shaping dominates all simpler mechanisms in every setting.

2.4 External Scenario-Seed Validation

Table 4 evaluates frozen models under external scenario seeds `test_start_seed=20000` and `30000`, aggregating 300 episodes per label at density 0.15. **shield_only** reaches 0.687 success with cost 0.207, while **proposed_gated_risk** reaches 0.663 success with cost 0.213. Their Wilson 95% success intervals overlap, so the correct claim is that gated risk is competitive and useful as a refinement, not that it is statistically superior to shield-only. The no-action-guard and risk-only controls fail in the same distinct

Method	Success $\uparrow$	95% CI	Cost $\downarrow$	Route $\uparrow$
Shield	0.687	[0.632, 0.737]	0.207	0.878
Gated risk	0.663	[0.608, 0.714]	0.213	0.866
Guard	0.613	[0.557, 0.667]	0.240	0.852
Retuned full	0.597	[0.540, 0.651]	0.280	0.838
Risk-only	0.000	[0.000, 0.013]	0.000	0.009
No guard	0.000	[0.000, 0.013]	1.000	0.191

Table 4: External scenario-seed validation at density 0.15. Each row aggregates 300 episodes from train seeds 0/1/2 and external test-start seeds 20000/30000. Confidence intervals are Wilson 95% intervals for success.

ways seen in the formal table: unsafe driving for no-action-guard and no-progress stagnation for risk-only.

2.5 Qualitative Behavior Evidence

The qualitative rollout package provides representative trajectory and behavior plots. These figures should be used to illustrate failure modes and mechanisms: no-action-guard tends toward high-speed collision or road departure; risk-only avoids cost by barely progressing; guard/shield-centered policies keep meaningful route progress while reducing unsafe events. These figures are explanatory and should not replace aggregate tables.

2.6 Claim Boundary

The result package supports three claim levels. First, a strong claim: action guard/shield is the dominant stabilizing mechanism in dense traffic. Second, a strong negative-control claim: risk-only and no-action-guard fail for different reasons, so low cost must be interpreted together with route completion. Third, a moderate refinement claim: TTC, lane, smoothness, and gated-risk shaping improve or explain behavior around the guard/shield mechanism, but current evidence does not justify presenting them as independently dominant.

This boundary is important for writing the discussion. The strongest paper is not a reward-only story; it is a mechanism-separated safe-RL story with formal, ablation, stress, external-seed, and qualitative evidence all pointing to the same guard/shield-centered interpretation.